Name: Asmit Pandey Roll No.2401072 Branch:AI&DS Class:T21 Subject:ASD&D

Experiment 1

Aim:

Study and Implementation of Basic Git and GitHub Operations.

This Experiment maps to LO2.

LO2: To understand and perform the basic version control operations using Git and GitHub, including repository creation, cloning, file tracking, committing, pushing, pulling, branching, merging, and remote repository management.

Theory:

Git is a distributed version control system that helps developers track changes in source code and files. It enables efficient project management through repositories, commits, branches, and remote repositories, making collaboration and version management easier. In this experiment, various Git commands are performed to understand the basic operations of version control.

Commands:

No.	Command	Purpose
1	git --version	Checks the installed Git version.
2	git config --global user.name "Archita Harchwani"	Sets the Git username globally.
3	git config --global user.email "email"	Sets the Git email globally.
4	git config --list	Displays Git configuration details.
5	mkdir gitproject	Creates a new project folder.
6	cd gitproject	Enters the project folder.
7	git init	Initializes a new Git repository.
8	echo "Hello Git" > file.txt	Creates a new file with content.
9	cat file.txt	Displays the contents of the file.
10	git status	Checks the current repository status.
11	git add file.txt	Adds a specific file to the staging area.
12	git add .	Adds all modified files to the staging area.
13	git commit -m "Initial commit"	Creates the first committed version.

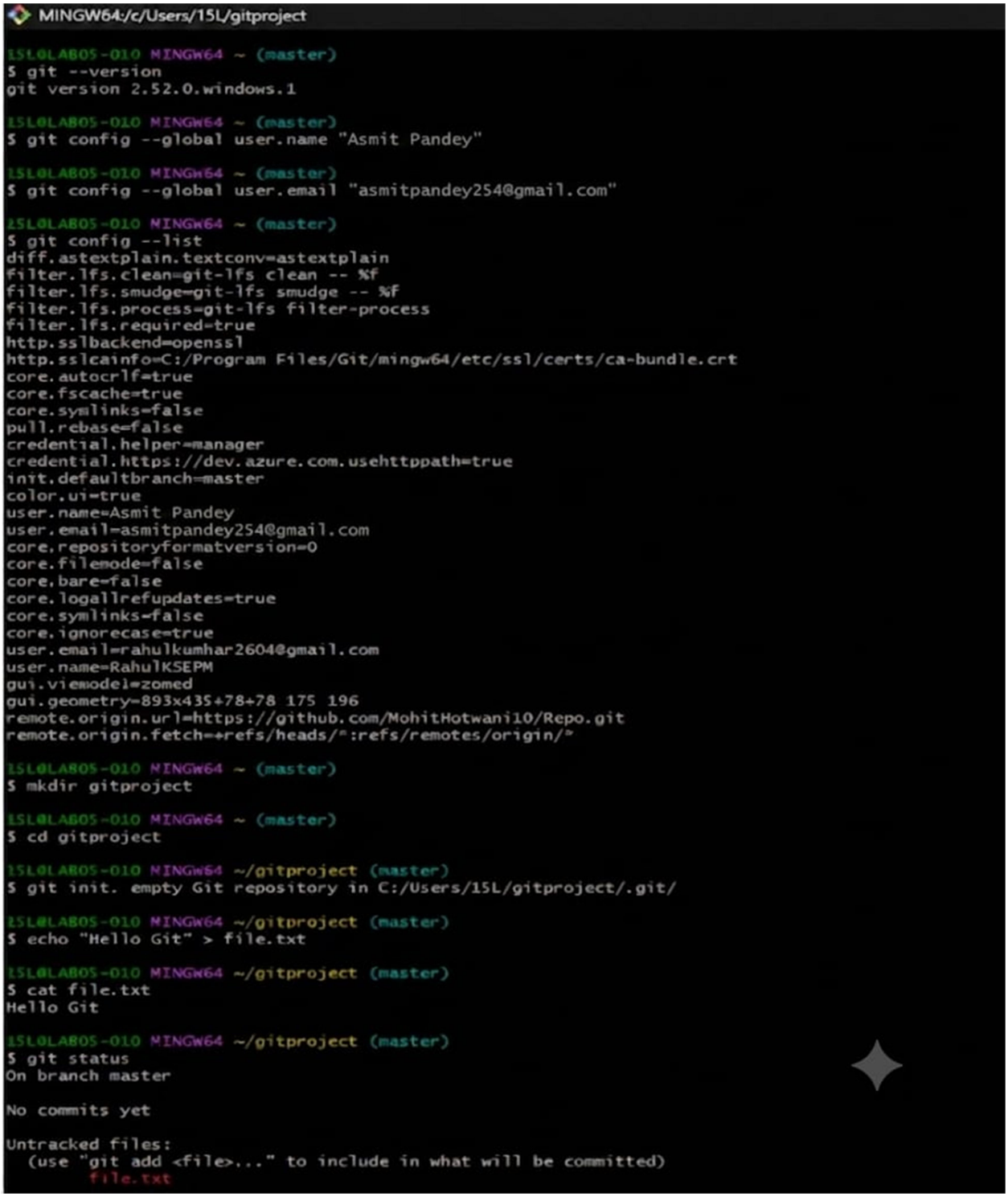
Name: Asmit Pandey Roll No.2401072 Branch:AI&DS Class:T21 Subject:ASD&D

No.	Command	Purpose
14	git log	Displays detailed commit/version history.
15	git log --oneline	Displays different versions using short commit IDs.
16	echo "Git Version Control" >> file.txt	Adds content to an existing file.
17	git diff	Displays changes between the current and committed file.
18	git restore file.txt	Rolls back uncommitted file changes.
19	git reset HEAD~1	Rolls back the latest commit while retaining changes.
20	git branch feature	Creates a new branch named feature.
21	git branch	Displays all branches and the current branch.
22	git checkout feature	Switches to the feature branch.
23	git switch feature	Switches to the feature branch using the modern Git command.
24	echo "Feature Branch" > feature.txt	Creates a file in the feature branch.
25	git checkout main	Switches back to the main branch.
26	git switch main	Switches back to the main branch using git switch.
27	git merge feature	Merges the feature branch with the current branch.
28	git branch -d feature	Deletes the feature branch.
29	git remote add origin <URL>	Connects the local repository to a remote repository.
30	git remote -v	Displays configured remote repositories.
31	git push -u origin main	Pushes local commits to the remote repository and authenticates if required.
32	git pull origin main	Retrieves and merges changes from the remote repository.
33	git clone <URL>	Creates a local copy of a remote repository.
34	git show <commit-id>:file.txt	Checks the content of a file in a specific version.
35	git checkout <commit-id>	Switches the repository to a previous committed

No.	Command	Purpose
		version.
36	git checkout main	Returns from an older version to the latest main branch.
37	git restore --source=<commit-id> file.txt	Restores a file from a specific older version.

Output:

Name: Asmit Pandey Roll No.2401072 Branch:AI&DS Class:T21 Subject:ASD&D



Name: Asmit Pandey Roll No.2401072 Branch:AI&DS Class:T21 Subject:ASD&D

```
15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git add file.txt
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next
 time Git touches it

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ ^C

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git add .

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git commit -m "Initial commit"
[master (root-commit) 38e9bca] Initial commit
 1 file changed, 1 insertion(+)
 create mode 100644 file.txt

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git log
commit 38e9bcab11b584a4823a0d92e6cdee2f003288169 (HEAD -> master)
Author: Asmit Pandey <asmitpandey254@gmail.com>
Date:   Wed Jul 15 13:34:12 2026 +0530

    Initial commit

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git log --oneline
38e9bca (HEAD -> master) Initial commit

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ echo "Git Version Control" >> file.txt

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ cat file.txt
Hello Git
Git Version Control

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git diff
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next
 Time Git touches it
diff --git a/file.txt b/file.txt
index 9f4d96d..e4fc519 100644
--- a/file.txt
+++ b/file.txt
@@ -1 +1,2 @@
 Hello Git
+Git Version Control

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git restore file.txt

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ cat file.txt
Hello Git

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git reset HEAD-1
fatal: ambiguous argument 'HEAD-1': unknown revision or path not in the working
tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git status
On branch master  file.txt
nothing to commit, working tree clean
```

Name: Asmit Pandey Roll No.2401072 Branch:AI&DS Class:T21 Subject:ASD&D

```
15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git log --oneline
38e9bca (HEAD -> master) Initial commit

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ echo "Second change" >> file.txt

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git add file.txt
warning: in the working copy of 'file.txt', LF will be replaced by CRLF the next
time Git touches it

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git commit -m "Second commit"
[master e7b43d2] Second commit
 1 file changed, 1 insertion(+)

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git reset HEAD~1
Unstaged changes after reset:
  M       file.txt

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ ^C

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   file.txt

no changes added to commit (use "git add" and/or "git commit -a")
15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git commit -m "Initial commit"
[master 6237020] Initial commit
 1 file changed, 1 insertion(+)

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git branch feature
15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git branch
  feature
* master
15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git checkout feature
Switched to branch 'feature'

15L@LAB05-010 MINGW64 ~/gitproject (feature)
$ git switch feature
Already on 'feature'

15L@LAB05-010 MINGW64 ~/gitproject (feature)
$ echo "Feature Branch" > feature.txt
```

```
feature
master

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git checkout feature
Switched to branch 'feature'

15L@LAB05-010 MINGW64 ~/gitproject (feature)
$ git switch feature
Already on 'feature'

15L@LAB05-010 MINGW64 ~/gitproject (feature)
$ echo "Feature Branch" > feature.txt

15L@LAB05-010 MINGW64 ~/gitproject (feature)
$ git add feature.txt
warning: in the working copy of 'feature.txt', LF will be replaced by CRLF the next
 time Git touches it

15L@LAB05-010 MINGW64 ~/gitproject (feature)
$ git commit -m "Added feature file"
[feature 97f7537] Added feature file
 1 file changed, 1 insertion(+)
 create mode 100644 feature.txt

15L@LAB05-010 MINGW64 ~/gitproject (feature)
$ git checkout main
error: pathspec 'main' did not match any file(s) known to git

15L@LAB05-010 MINGW64 ~/gitproject (feature)
$ git checkout master
Switched to branch 'master'

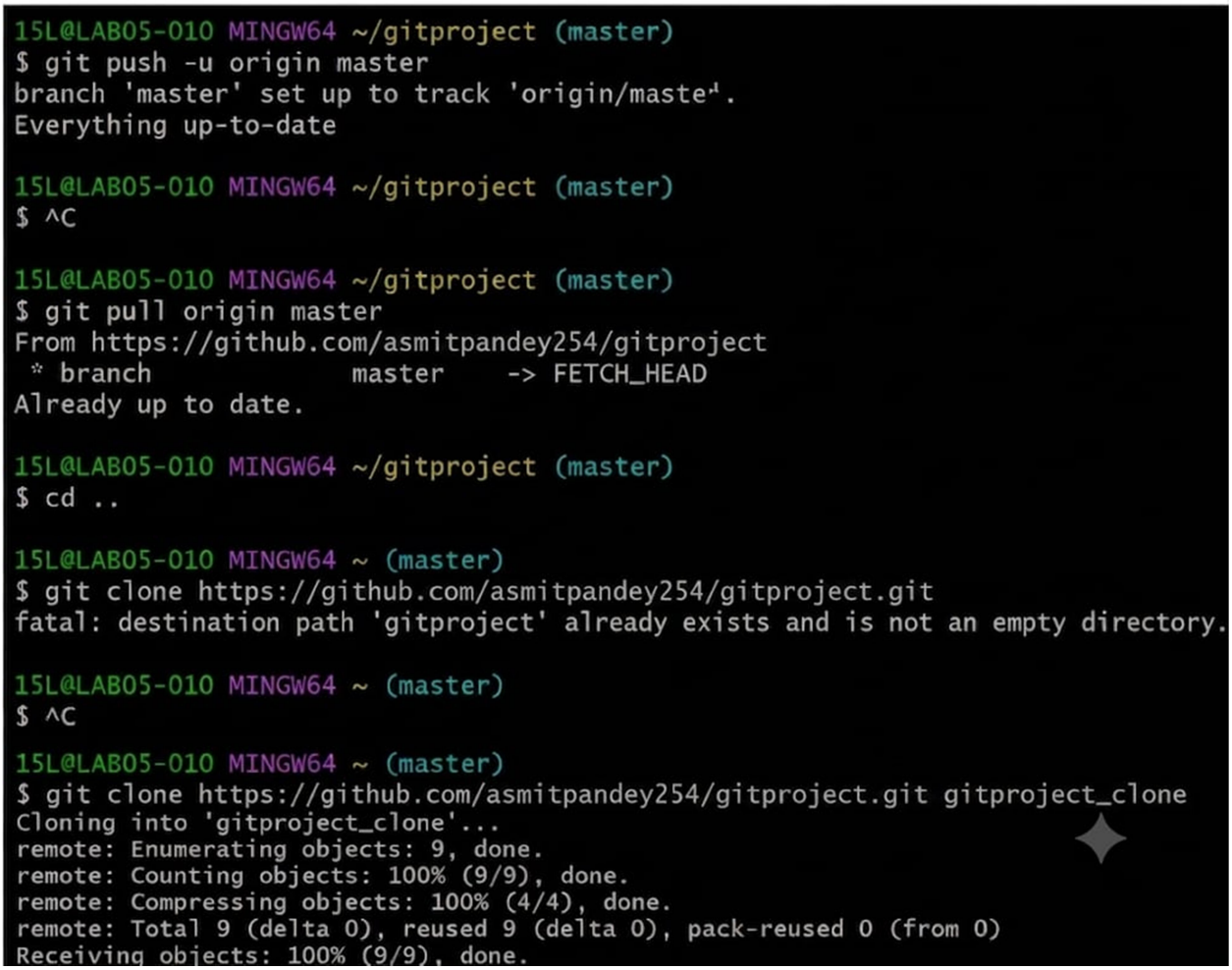
15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git switch master
Already on 'master'

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git merge feature
Updating 6237020..97f7537
Fast-forward
 feature.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 feature.txt

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ ^C

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git branch -d feature
Deleted branch feature (was 97f7537).
```

Name: Asmit Pandey Roll No.2401072 Branch:AI&DS Class:T21 Subject:ASD&D



Name: Asmit Pandey Roll No.2401072 Branch:AI&DS Class:T21 Subject:ASD&D

```
15L@LAB05-010 MINGW64 ~(master)
$ cd ~/gitproject

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ cd /c/Users/15L/gitproject

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ pwd
/c/Users/15L/gitproject

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git log --oneline
97f7537 (HEAD -> master, origin/master) Added feature file
6237020 Initial commit
38e9bca Initial commit

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ ls
feature.txt  file.txt

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ ^C

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git show 6237020:file.txt
Hello Git
Second change

15L@LAB05-010 MINGW64 ~/gitproject (master)
$ git checkout 6237020
Note: switching to '6237020'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

   git switch -c <new-branch-name>

Or undo this operation with:

   git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at 6237020 Initial commit
15L@LAB05-010 MINGW64 ~/gitproject ((6237020...))
$ git checkout master
Previous HEAD position was 6237020 Initial commit
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
```

Conclusion:

The experiment was successfully completed using basic Git commands to create and manage a repository, track changes, perform commits, manage branches, and work with a remote GitHub repository. It helped in understanding the fundamentals of Git and version control.